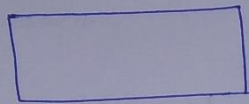


Symbols:

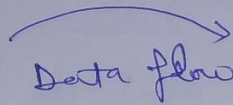
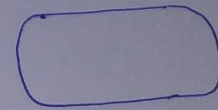


External entity
(Input/output)

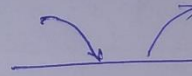


process/Function

or



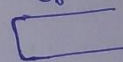
Data flow



or

Data store

or



(Repository of data ; the arrows
indicate net input and net
output to store)

Leveling:- DFD represents a system or software at any level of abstraction.

④ Level 0 DFD:- A Level 0 DFD, also called a fundamental system model or context diagram represents the entire S/W element as a single bubble with input and output data indicated by incoming and outgoing arrows respectively.

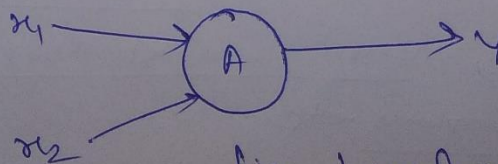
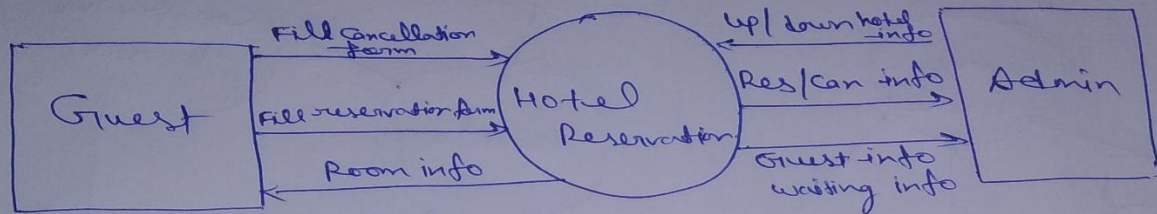


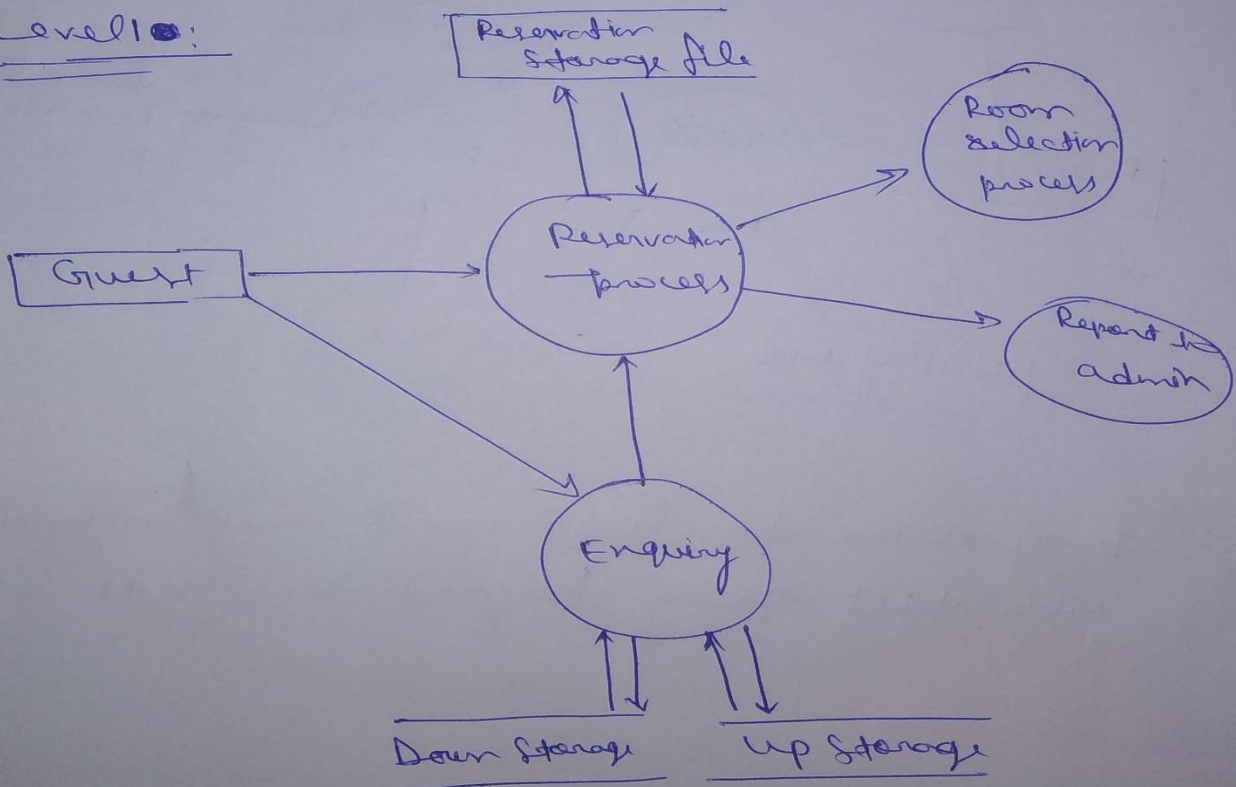
Fig: Level-0 DFD

Ex: Hotel Reservation System

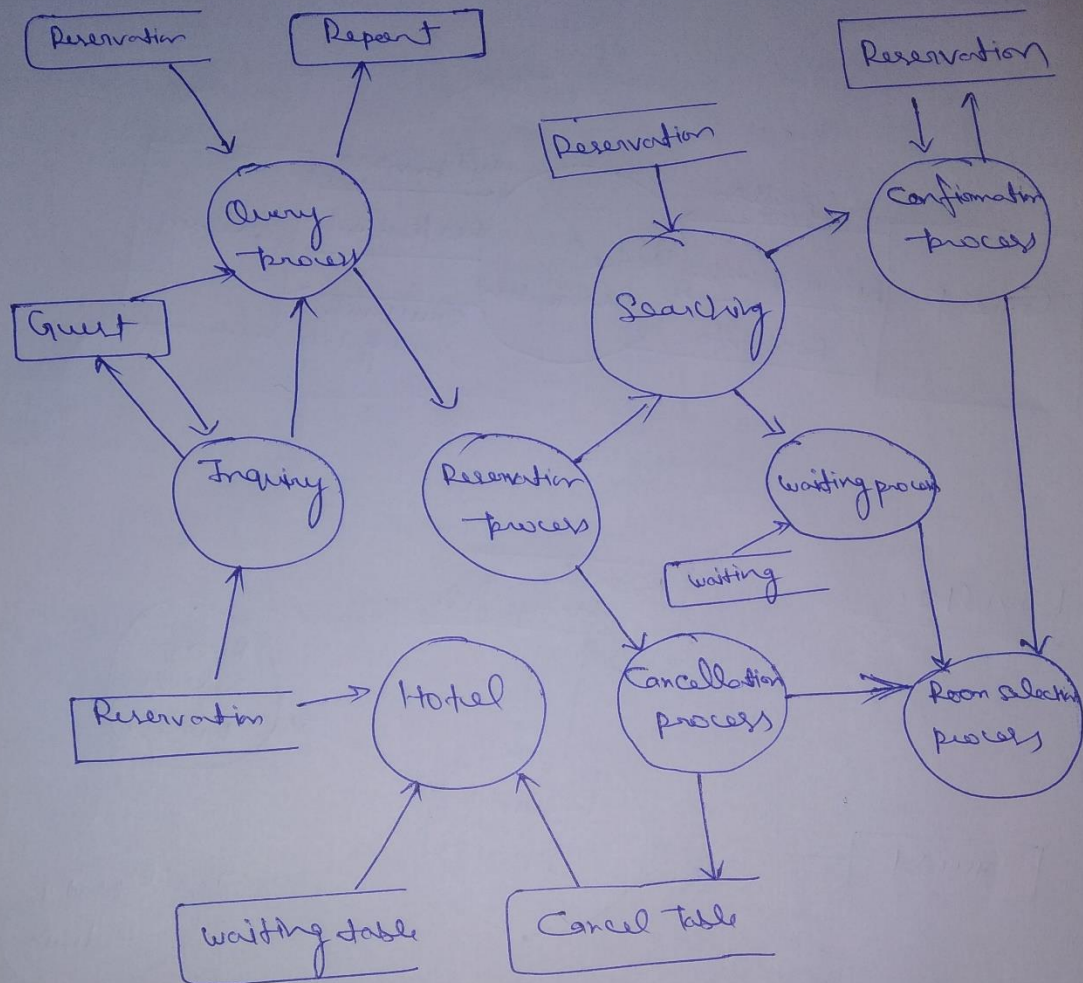
Level 0: OR Context diagram



Level 1:

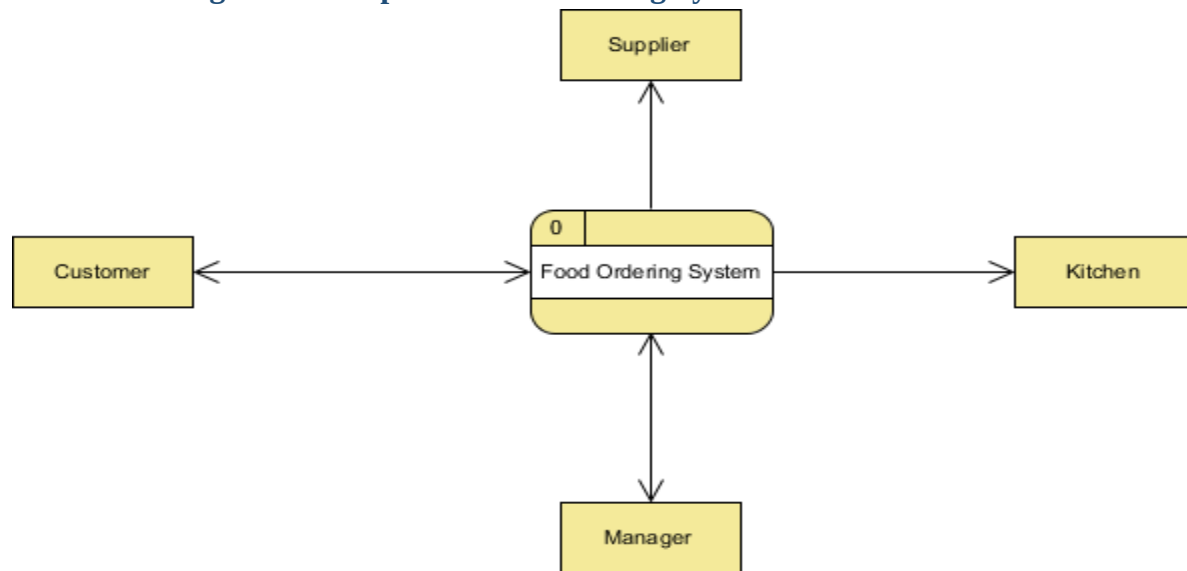


Level 2 DFD :-

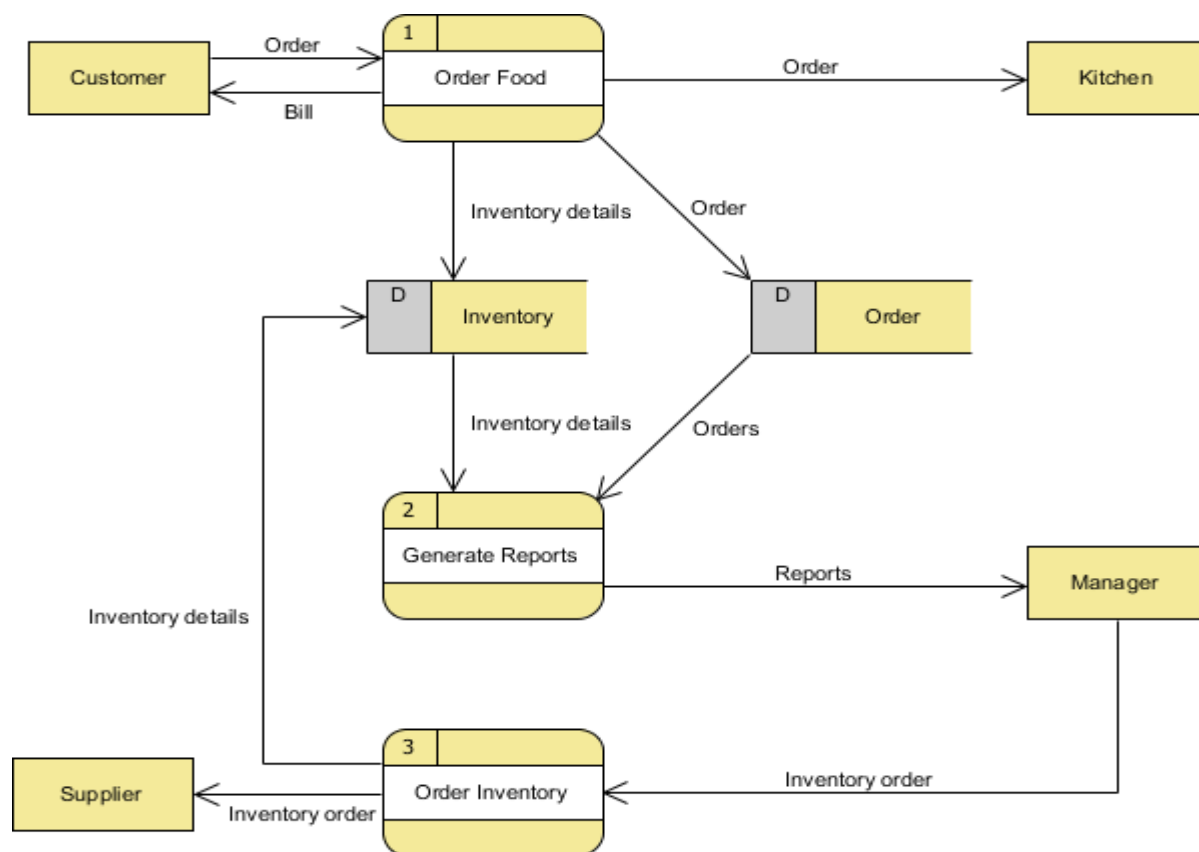


Note: Progression to level 3 or 4 and beyond is possible, but going beyond level 3 is uncommon.

Data Flow Diagram: Examples - Food Ordering System: Level 0 DFD



Level 1 DFD



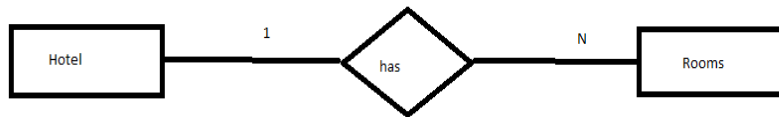
(b) Data Dictionary: are simply repositories to store information about all data items defined in DFD. A data dictionary contains metadata i.e data about the database.

DATA					DATA DICTIONARY (METADATA)		
employee_id	first_name	last_name	nin	department_id	Column	Data Type	Description
44	Simon	Martinez	HH 45 09 73 D	1	employee_id	int	Primary key of a table
45	Thomas	Goldstein	SA 75 35 42 B	2	first_name	nvarchar(50)	Employee first name
46	Eugene	Comelsen	NE 22 63 82	2	last_name	nvarchar(50)	Employee last name
47	Andrew	Petculescu	XY 29 87 61 A	1	nin	nvarchar(15)	National Identification Number
48	Ruth	Stadick	MA 12 89 36 A	15	position	nvarchar(50)	Current position title, e.g. Secretary
49	Barry	Scardelis	AT 20 73 18	2	department_id	int	Employee department. Ref: Departments
50	Sidney	Hunter	HW 12 94 21 C	6	gender	char(1)	M = Male, F = Female, Null = unknown
51	Jeffrey	Evans	LX 13 26 39 B	6	employment_start_date	date	Start date of employment in organization.
52	Doris	Berndt	YA 49 88 11 A	3	employment_end_date	date	Employment end date. Null if employee still employed
53	Diane	Eaton	BE 08 74 68 A	1			
54	Bonnie	Hall	WW 53 77 68 A	15			
55	Taylor	Li	ZE 55 22 80 B	1			

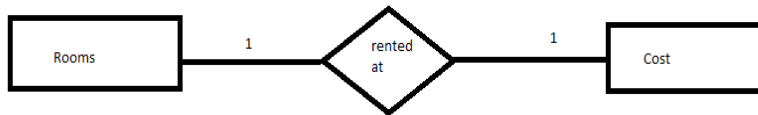
Data Dictionary

Data Dictionary outlining a Database on Driver Details in NSW

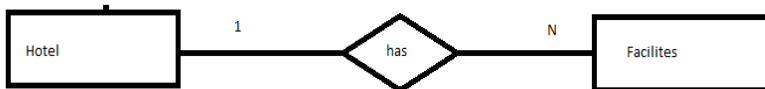
Field Name	Data Type	Data Format	Field Size	Description	Example
License ID	Integer	NNNNNN	6	Unique number ID for all drivers	12345
Surname	Text		20	Surname for Driver	Jones
First Name	Text		20	First Name for Driver	Arnold
Address	Text		50	First Name for Driver	11 Rocky st Como 2233
Phone No.	Text		10	License holders contact number	0400111222
D.O.B	Date / Time	DD/MM/YYYY	10	Drivers Date of Birth	08/05/1956



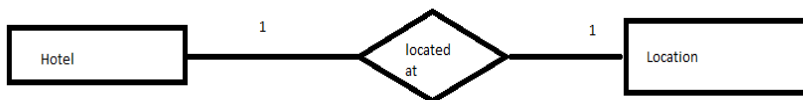
Each hotel provides many rooms.



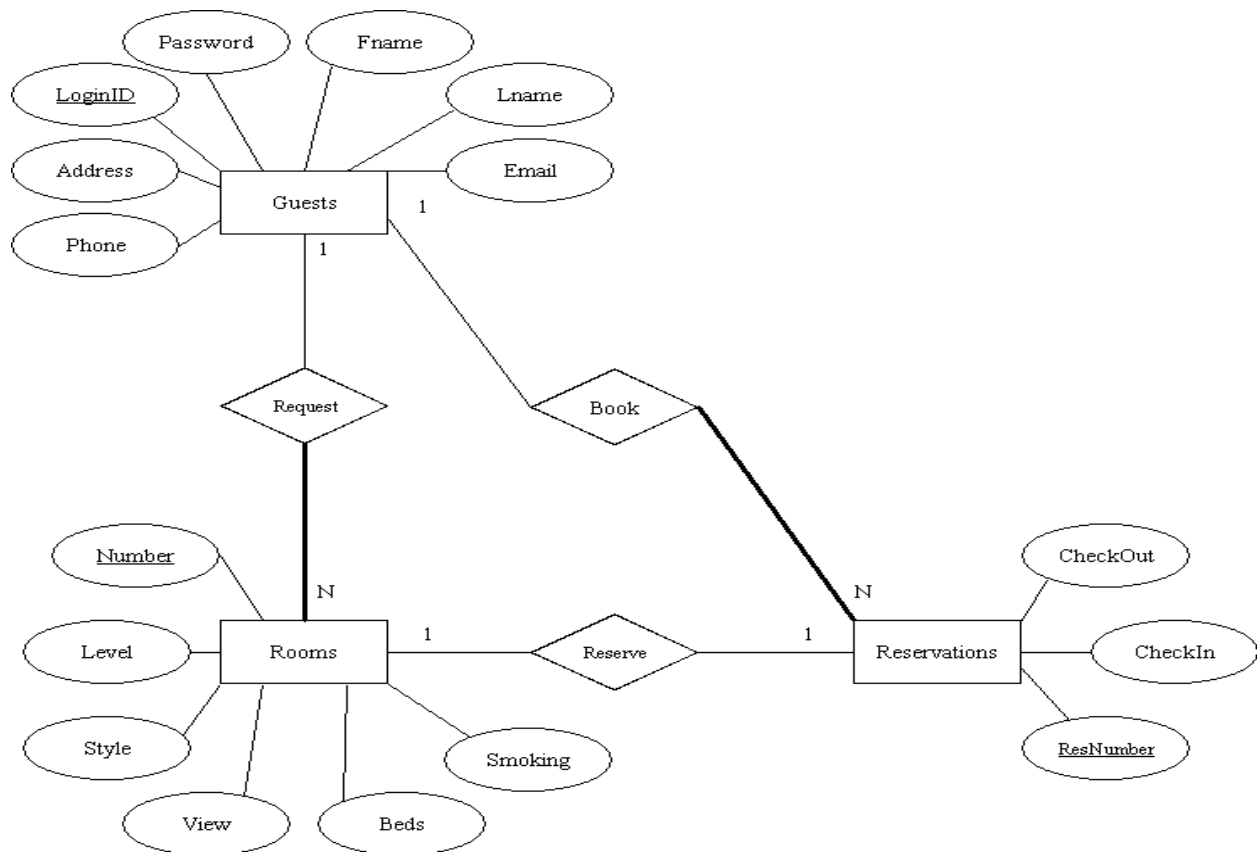
Each room is rented at a cost.



A hotel has many facilities available with that.



A hotel is located at a particular location.



ER Diagram ---> Hotel Reservation System

(d) Decision Table: A decision table is used to represent the complex processing logic in a tabular or a matrix form. The upper rows of the table specify the variables or conditions to be evaluated. The lower rows of the table specify the actions to be taken when the corresponding conditions are satisfied. A column in a table is called a rule. A rule implies that if a condition is true, then the corresponding action is to be executed.

Conditions

Valid selection	No	Yes	Yes	Yes
New member	-	Yes	No	No
Renewal	-	No	Yes	No
Cancellation	-	No	No	Yes
Actions				
Display error message	X	-	-	-
Ask member's details	-	X	-	-
Build customer record	-	X	-	-
Generate bill	-	X	X	-
Ask member's name & membership number	-	-	X	X
Update expiry date	-	-	X	-
Print cheque	-	-	-	X
Delete record	-	-	-	X

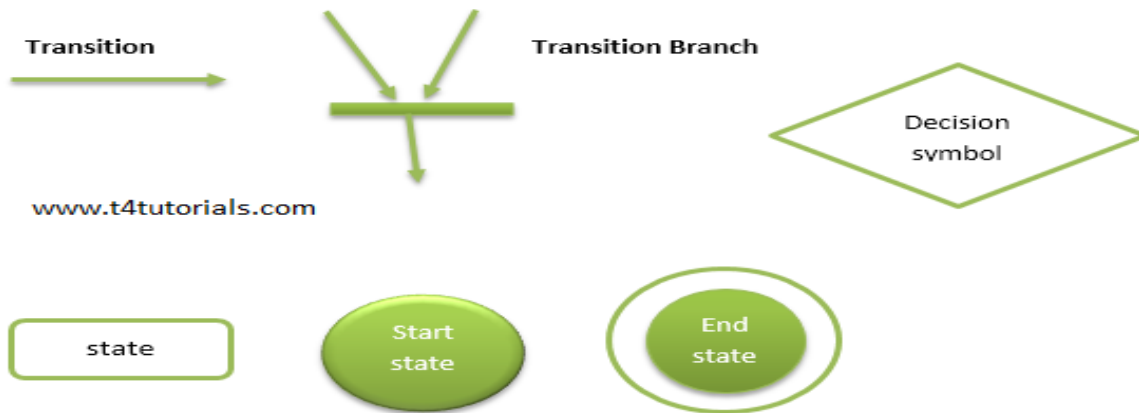
Decision table for LMS

From the above table you can easily understand that, if the valid selection condition is false then the action taken for this condition is 'display error message'. Similarly, the actions taken for other conditions can be inferred from the table.

(e) State Transition diagrams:

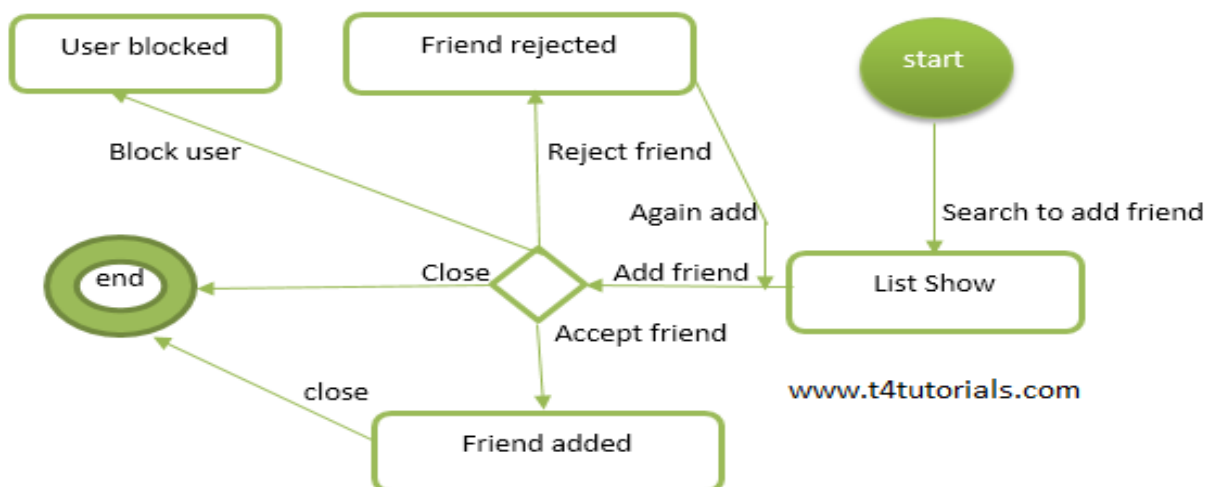
When we write our program in Object-oriented programming, then we need to make classes and objects. Objects change its states during different functions/actions performed on objects.

Symbols:



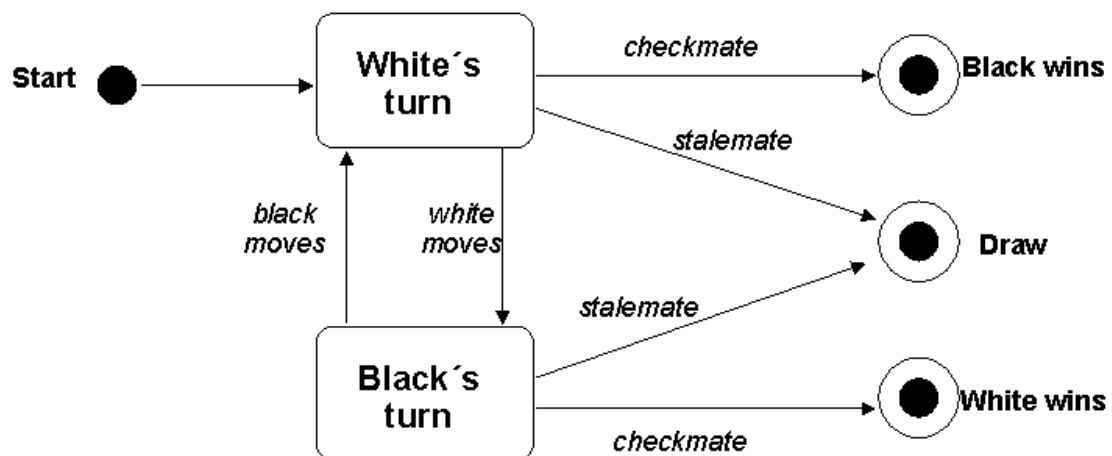
Example of State Transition Diagram:

Case Study: You need to develop a web-based application in such a way that user can search other users, and after getting search complete, the user can send the friend request to other users. If the request is accepted, then both users are added to the friend list of each other. If one user does not accept the friend request. The second user can send another friend request. The user can also block each other.



UML State Diagram - example

Chess game



1. Introduction

Describe a quick background of the use case.

2. Actors

List the actors that interact and participate in the use cases.

3. Pre Conditions

Pre conditions that need to be satisfied for the use case to perform.

4. Post Conditions

Define the different states in which we expect the system to be in, after the use case executes.

5. Flow of events

1. Basic Flow

List the primary events that will occur when this use case is executed.

1. Alternative Flows

Any Subsidiary events that can occur in the use case should be separately listed. List each such event as an alternative flow. A use case can have many alternative flows as required.

6. Special Requirements

Business rules should be listed for basic & information flows as special requirements in the use case narration. These rules will also be used for writing test cases. Both success and failures scenarios should be described.

7. Use Case relationships

For Complex systems it is recommended to document the relationships between use cases. Listing the relationships between use cases also provides a mechanism for traceability.

Use Case Template